



ATARI BREAKOUT DOCUMENTATION



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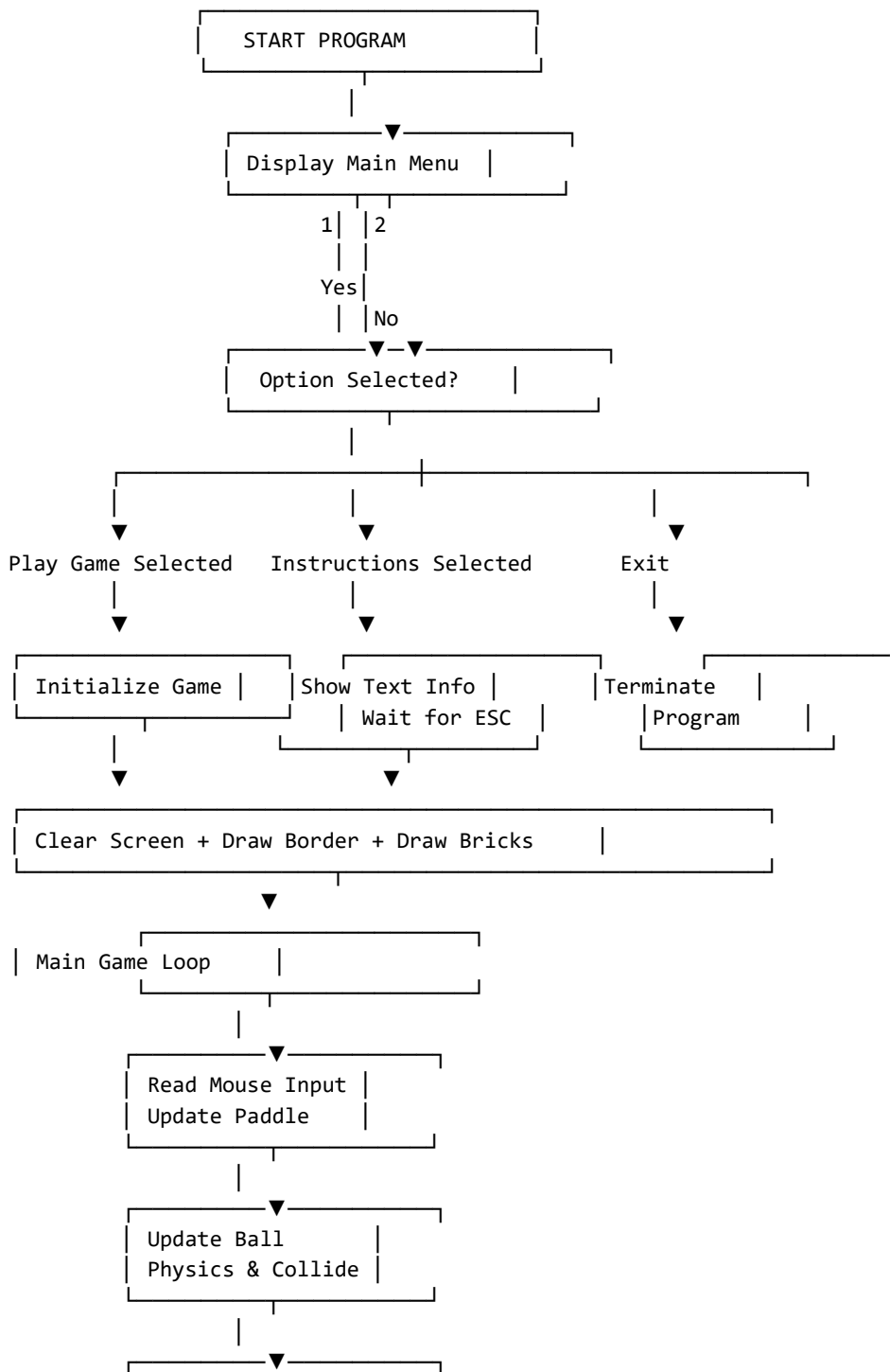
To: Sir Talha Arif

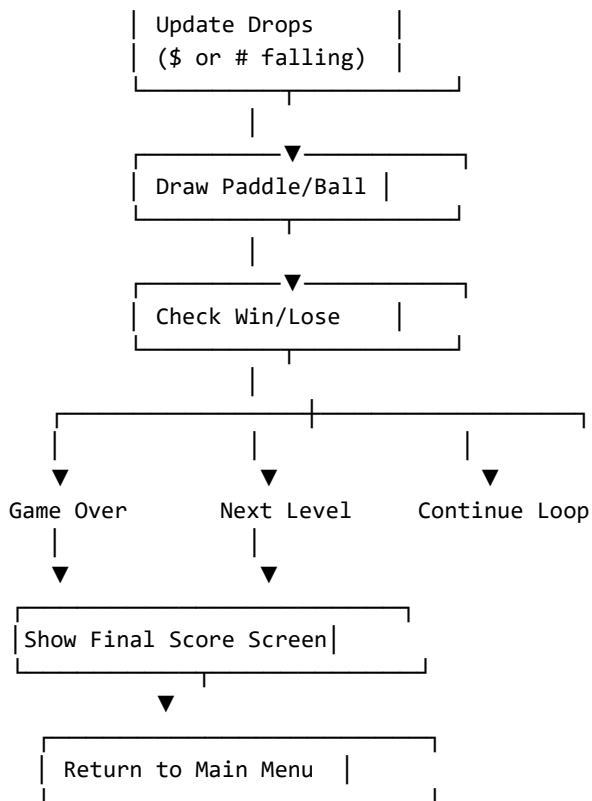
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1. FLOWCHART OF GAME LOGIC





2. INTERRUPT USAGE

Program uses three main interrupt families:

2.1 Mouse Interrupt — INT 33h

```
mov ax, 3 int 0x33
```

Reads mouse X/Y coordinates and updates paddle position.

2.2 Keyboard Interrupt — INT 16h

```
mov ah, 01 int 16h
```

Used for menu navigation and ESC detection.

2.3 PIT & Speaker Control

```
mov al, 0xB6 out
0x43, al out
0x42, al out
0x42, ah
```

Used for sound effects (brick hit, paddle hit, death sound).

3. SCORING SYSTEM & GAMEPLAY MECHANICS

3.1 Lives

- Player starts with 3 lives
- Lose life when ball falls or bomb hits
- 0 lives → Game Over

3.2 Scoring

- Normal bricks: 1 HP
- Red bricks: 2 HP
- Drops: \$ = +20 points, # = lose 1 life

3.3 Levels

Level 1: basic bricks

Level 2: bricks + falling drops

3.4 Paddle Mechanics

Follows mouse movement; restricted to boundaries.

3.5 Ball Physics

`ballX += ballDX`

`ballY += ballDY`

Collision with walls, bricks, paddle, bottom boundary.

3.6 Game Over Conditions

- lives = 0 → lose
- Level 2 cleared → win

4. ADDITIONAL NOTES

- Text buffer = 0xB800
- Brick HP stored in bricks[60]
- Values: 0, 1, 2